A Constructive Introduction to Quantum Field Theory

The Polchinski flow and forward–backward equations

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SISSA lectures, Lecture 4 (provisional) — notes: Lecture 4, §§7–11

Where we are

So far. Stochastic quantisation of the GFF; the field at resolution t is a martingale of smooth fields; **observables** are martingales in the filtration of finite-resolution measurements.

Today. Interactions, still indexed by the **resolution** $t \in [0, \infty)$.

1. Interactions: the **Polchinski equation** and its **FBSDE**; the force is an observable.
2. The FBSDE in force coordinates $\Rightarrow$ the **Wetterich equation**.
3. Variational structure: Boué–Dupuis, HJB, infinite volume.
4. Additional material: flatness, counterterms for Φ^4_3, DS equations at every scale.

Caveat. Statements marked (formal) are exact in finite dimensions, with a terminal resolution $T < \infty$; the limits $T \to \infty$, $L \to \infty$ are the real problems. Bailleul–Chevyrev–G. (in prep.); Polchinski 1984; Bauerschmidt–Bodineau–Dagallier 2024

Interactions: the Polchinski equation and FBSDEs

Lecture 4

Effective potentials

Finite volume $\mathbb{T}_L^d$, terminal resolution $T < \infty$ (UV cutoff), interaction V_T bounded below, e.g.

$$V_T(\varphi) = \int_{\mathbb{T}_L^d} [\lambda H_4(\varphi; c_T) + \gamma_T H_2(\varphi; c_T)], \qquad \nu_T(\mathrm{d}\varphi) = Z_T^{-1} \mathrm{e}^{-V_T(\varphi)} \mu^{Q_T}(\mathrm{d}\varphi).$$

Definition. (Wilson effective action)

$$\mathrm{e}^{-V_t(\varphi)} := \int \mathrm{e}^{-V_T(\varphi+\zeta)} \, \mu^{Q_{T,t}}(\mathrm{d}\zeta), \qquad t \in [0, T].$$

The interaction felt at resolution t once the scales $[t, T]$ are integrated out; $V_0(0) = -\log Z_T$ is the free energy.

$$e^{-V_t(X_t)} = \mathbb{E}_t[\mathrm{e}^{-V_T(X_T)}]$$

Marginals and stochastic localisation

Proposition. (Marginals) $\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{P}_0 = Z_T^{-1}\mathrm{e}^{-V_T(X_T)}$. *Then* $\mathrm{Law}_{\mathbb{P}}(X_t) = \nu_t \propto \mathrm{e}^{-V_t}\mu^{Q_t}$, *and* $\mathrm{Law}_{\mathbb{P}}(X_T|\mathscr{F}_t) = \mu_t^{X_t}$, *the tilt of* $\mu^{Q_{T,t}}(\cdot - X_t)$ *by* e^{-V_T}.

- At every resolution the field is again a **Gibbs measure**: reference Q_t, potential V_t.
- The measure-valued martingale $t \mapsto \mu_t^{X_t}$ goes from ν_T ($t=0$) to δ_{X_T} ($t=T$): a **stochastic localisation** of ν_T (Eldan), adapted to the covariance profile.

Proof of the marginals. Under $\mathbb{P}_0$, $X_T = X_t + (X_T - X_t)$ with independent increment $\sim\mu^{Q_{T,t}}$, so

$$\mathbb{E}_{\mathbb{P}}[G(X_t)] = Z_T^{-1}\,\mathbb{E}_0[G(X_t)\mathrm{e}^{-V_T(X_T)}] = Z_T^{-1}\,\mathbb{E}_0[G(X_t)\mathrm{e}^{-V_t(X_t)}].$$

The Polchinski equation

Theorem. (Polchinski 1984) (formal)

$$\partial_t V_t = \tfrac{1}{2}\langle D V_t, \dot{Q}_t \, D V_t\rangle - \tfrac{1}{2}\mathrm{Tr}(\dot{Q}_t \, D^2 V_t), \qquad V_T \textit{given}.$$

Proof. $u_t = \mathrm{e}^{-V_t} = \mathrm{e}^{\frac{1}{2}\Delta_{Q_T,t}} u_T$, $\Delta_Q = \mathrm{Tr}(Q D^2)$, solves the backward heat equation $\partial_t u_t = -\frac{1}{2}\mathrm{Tr}(\dot{Q}_t D^2 u_t)$. Insert $D^2 u = (D V \otimes D V - D^2 V)\, u$.

Nonlinear term: generates new, nonlocal interactions: **trees**.

Trace term: contracts pairs of fields with the shell covariance $\dot{Q}_t \, \mathrm{d}t$: **loops**.

A backward Hamilton–Jacobi equation; $u = \mathrm{e}^{-V}$ is a Cole–Hopf transform.

The Polchinski process

Theorem. (FBSDE for the interacting field) (formal) *Under* $\mathbb{P}$*, the* ***interacting field at resolution*** t $\Phi_t := X_t$ *solves*

$$\mathrm{d}\Phi_t = \dot{Q}_t\, F_t\, \mathrm{d}t + \dot{Q}_t^{1/2}\, \mathrm{d}W_t^{\mathbb{P}}, \qquad F_t = -D\,V_t(\Phi_t) = \mathbb{E}_t^{\mathbb{P}}\left[-D\,V_T(\Phi_T)\right].$$

The ***effective force*** F *is a* $\mathbb{P}$*-martingale: an observable field of the interacting theory.*

Proof. Density process $M_t = Z_T^{-1} \mathrm{e}^{-V_t(X_t)}$. Itô + Polchinski: no drift, $\mathrm{d}M_t = -M_t \langle D\,V_t(X_t), \dot{Q}_t^{1/2} \mathrm{d}W_t \rangle$. Girsanov: $W_t^{\mathbb{P}} = W_t + \int_0^t \dot{Q}_s^{1/2}\, D\,V_s(X_s)\, \mathrm{d}s$. Finally $D\,V_t(\phi) = \mu_t^{\phi}\,(D\,V_T)$.

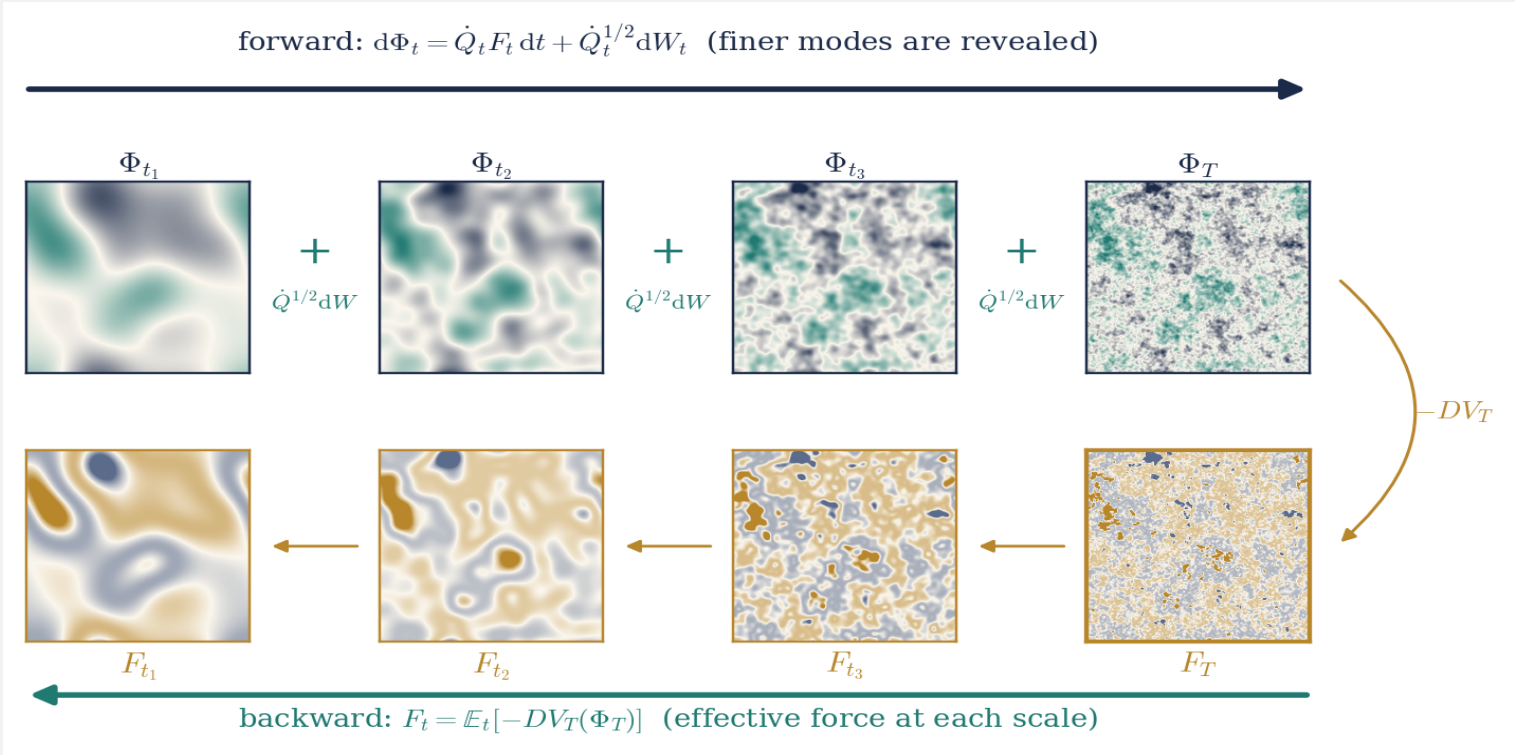
A forward–backward system

Read as an equation for an unknown pair (Φ, F):

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\left[-D\,V_T(\Phi_T)\right], \qquad t \in [0,T].$$

- The field is driven **forward** by a force determined **backward** from the terminal field.
- No Polchinski equation needed; $\nu_T = \mathrm{Law}(\Phi_T)$ from the solution.
- **Strategy**: solve with renormalised V_T, estimates uniform in T and volume, take limits.
- Φ is **not** an observable (a semimartingale); observables are built from the noise and from F.

Forward field, backward force



$\mathbb{T}^2$, heat profile, same noise. The force is shown to first order in λ: $F_t = -4\lambda H_3(\Phi_t; c_t)$, each panel in its own units.

Closed versus local

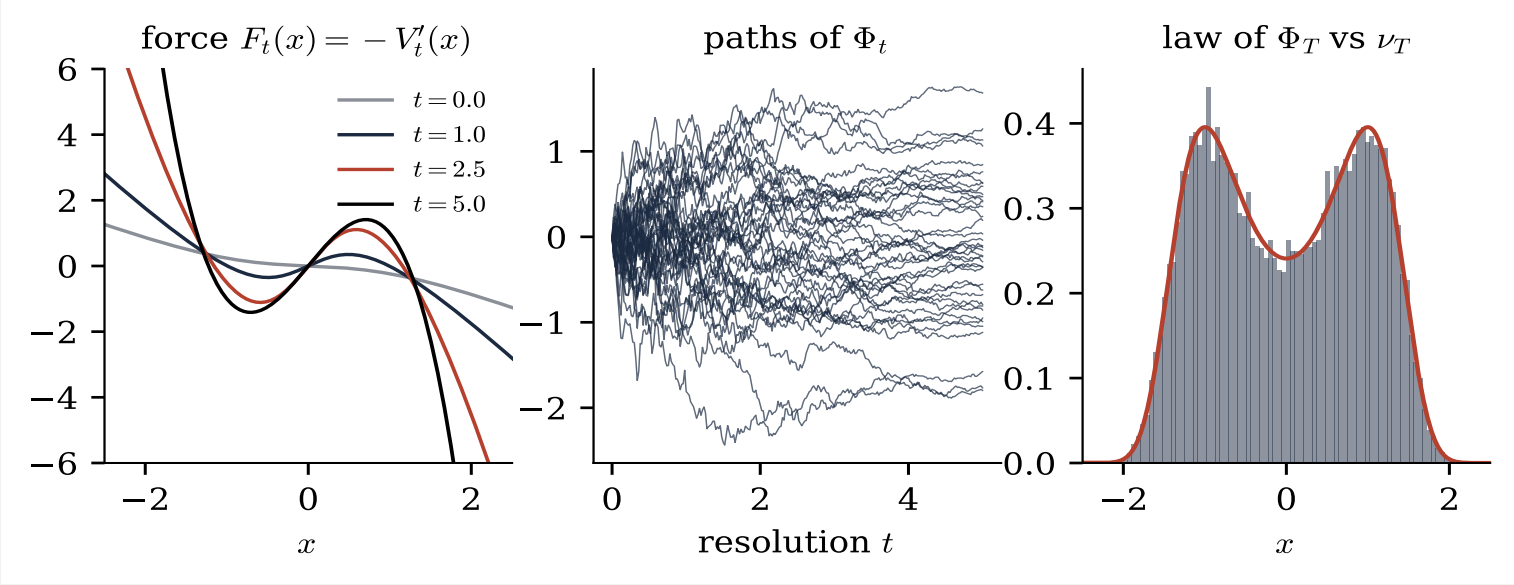
At finite T the force is a **closed** observable, $F_t = \mathbb{E}_t[F_T]$, and for Φ^4 it is **local**:

$$F_T(x) = -4\lambda H_3(\Phi_T(x); c_T) - 2\gamma_T H_1(\Phi_T(x); c_T).$$

As $T \to \infty$

F should converge **as an observable**. In $d = 2$ it is closed, $F_\infty = -4\lambda{:}\Phi^3_\infty{:}$. In $d = 3$ a closed limit $F_t = \mathbb{E}_t F_\infty$ would let the DS identity pass to $t \to \infty$, and conditioning on $\sigma(\Phi_\infty)$ would give an integrable logarithmic derivative of Φ^4_3 as a function of the field: not expected (Albeverio–Liang–Zegarlinski 2006: the regularised ones do not converge in L^2). Locality of the limiting force is the crux of the construction.

Zero dimensions



$Q_t = 1 - \mathrm{e}^{-t}$, $T = 5$, $V_T(x) = \frac{1}{2}x^4 - \frac{3}{2}x^2$. Left: $F_t = -V_t'$, almost linear at low resolution, double well as $t \to T$. Centre: FBSDE paths split between the wells (stochastic localisation). Right: histogram of Φ_T vs. the density of ν_T.

Observables of the interacting theory

For G a functional of the terminal field, $O_t = \mathbb{E}_t^{\mathbb{P}}[G(\Phi_T)] = G_t(\Phi_t)$, $G_t(\phi) = \mu_t^{\phi}(G)$, and

$$\partial_t G_t + \tfrac{1}{2}\mathsf{Tr}(\dot{Q}_t D^2 G_t) - \langle \dot{Q}_t D V_t, D G_t \rangle = 0, \qquad G_T = G:$$

the backward Kolmogorov equation, a Doob transform of the Gaussian heat equation. Any solution gives an observable, closed at finite T, not necessarily as $T \to \infty$. The force is $G = -D V_T$.

Tangent observables. V_T^{θ} depending on a coupling, $U_t = \partial_\theta V_t^{\theta}$:

$$U_t(\Phi_t) = \mathbb{E}_t^{\theta}[\partial_\theta V_T^{\theta}(\Phi_T)], \qquad \partial_\theta \mathbb{E}^{\nu_T^{\theta}}[G] = -\mathsf{Cov}^{\nu_T^{\theta}}(G, \partial_\theta V_T^{\theta}).$$

E.g. the running $:\varphi^4:$: for $\theta = \lambda$: observables **conjugate to the coordinates of theory space**.

The force: a Burgers equation

Differentiating Polchinski, $F_t = -D V_t$ solves the backward viscous **Burgers equation**

$$\partial_t F_t = -D F_t[\dot{Q}_t F_t] - \tfrac{1}{2}\mathsf{Tr}(\dot{Q}_t D^2 F_t), \qquad F_T = -D V_T.$$

- Cole–Hopf $F = D\log u$ linearises it ($u = \mathrm{e}^{-V}$ solves a heat equation).
- Uses only the force: **intensive**, while V_t is extensive ($\sim$ volume). Matters in infinite volume.
- Flows for different covariances commute when DF is symmetric, i.e. F a gradient. Non-gradient forces (general Wilson–Itô diffusions): curvature.

The FBSDE in force coordinates: the Wetterich equation

Lecture 4

Pontryagin's maximum principle

Minimise $J(u) = g(x_T) + \int_t^T \ell(x_s, u_s)\,\mathrm{d}s$ subject to $\dot{x} = b(x,u)$, $x_t = x$. **Hamiltonian:**

$$H(x,p,u) = \langle p, b(x,u)\rangle - \ell(x,u).$$

Pontryagin. Along an optimal pair there is a **costate** p with

$$\dot{x} = \partial_p H, \qquad \dot{p} = -\partial_x H, \qquad p_T = -Dg(x_T), \qquad u \in \arg\max H(x,p,\cdot).$$

- State forward, costate backward: a **two-point boundary problem**.
- The costate is the Lagrange multiplier of the constraint $\dot{x} = b$; an optimal control is a critical point of the Hamiltonian.
- Next: why this holds, and what p is.

Pontryagin–Boltyanskii–Gamkrelidze–Mishchenko 1962.

Pontryagin: first variation and HJB

First variation. Perturb $u \to u + \varepsilon v$; the state varies by $\xi = \partial_\varepsilon x$,

$$\dot{\xi} = \partial_x b\, \xi + \partial_u b\, v, \qquad \xi_t = 0.$$

The costate equation $\dot{p} = -\partial_x H$ is exactly what makes $\frac{\mathrm{d}}{\mathrm{d}s}\langle p, \xi\rangle = \langle \partial_x \ell, \xi\rangle + \langle p, \partial_u b\, v\rangle$. Integrating, with $p_T = -Dg(x_T)$:

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(u + \varepsilon v)|_{\varepsilon=0} = -\int_t^T \langle \partial_u H(x_s, p_s, u_s), v_s\rangle \,\mathrm{d}s.$$

HJB. Dynamic programming over $[t, t + \mathrm{d}t]$:

$$V(t,x) = \min_u \left[\ell\, \mathrm{d}t + V(t + \mathrm{d}t, x + b\,\mathrm{d}t)\right] \quad \Longrightarrow \quad \partial_t V = H_*(x, -DV), \quad H_* = \max_u H.$$

Hamilton's equations for H_* are its characteristics, and along them $p_s = -DV(s, x_s)$: the costate is the force the future exerts on the present.

The Hamiltonian of the field

State = field, $b = \dot{Q}_s^{1/2} u$, $\ell = \frac{1}{2}\|u\|^2$, $g = V_T$. Maximising over $u = \dot{Q}_s^{1/2} p$:

$$H_*(p) = \frac{1}{2}\langle p, \dot{Q}_s p\rangle:$$

a free particle with mass matrix $\dot{Q}_s^{-1}$, and no dependence on the state x.

- **Without noise:** $\dot{p} = 0$, $\dot{x} = \dot{Q}\, p$: the shear $(x, p) \mapsto (x + S_t p, p)$. All interaction is in $p = -D\, V_T(x + S_t p)$: **trees**. HJB = Polchinski without the trace.
- **With noise** (Bismut 1973, Peng 1990): the costate solves a BSDE

$$\mathrm{d}p_s = -\partial_x H\, \mathrm{d}s + \Upsilon_s\, \mathrm{d}W_s,$$

  with $\Upsilon\, \mathrm{d}W$ the price of adaptedness. Here $\partial_x H = 0$: the costate is a **martingale**, $p_t = -D\, V_t(X_t) = F_t$.
- The Itô term turns HJB into **Polchinski**: quadratic part = trees, trace term = loops.

The FBSDE is the stochastic Hamiltonian system: field = state, force = costate.

State and costate

$S_t := Q_T - Q_t$: the covariance still unresolved; $\dot{S}_t = -\dot{Q}_t$, $S_T = 0$. $\mathrm{d}F_t = \mathrm{Y}_t\,\mathrm{d}W_t$.

Control $\mathrm{d}X = \dot{Q}_t^{1/2}\,(u\,\mathrm{d}t + \mathrm{d}W)$, optimal $u_t = \dot{Q}_t^{1/2} p_t$, costate $p_t = F_t$.

The field Φ is the state; the force F is the costate.

- $F_t(\cdot)$ is the **decoupling field**, Burgers its quasilinear PDE.
- Transport term $-DF\,[\dot{Q}F]$ ← drift of the state: trees.
  Itô term $-\frac{1}{2}\mathsf{Tr}(\dot{Q}D^2F)$ ← noise: loops.

Noiseless flow from t to T: the linear **shear** Σ_t: $(x,p) \mapsto (x + S_t p,\, p)$, the characteristics of the transport part.

Removing the predictable drift: the forecast

Proposition. (formal) *The conditional mean of the terminal field is*

$$\psi_t := \mathbb{E}_t[\Phi_T] = \Phi_t + S_t F_t, \qquad (\psi_t, F_t) = \Sigma_t(\Phi_t, F_t),$$

a martingale with $\mathrm{d}\psi_t = \dot{Q}_t^{1/2}\mathrm{d}W_t + S_t\,\mathrm{d}F_t = (1 + S_t D F_t(\Phi_t))\,\dot{Q}_t^{1/2}\mathrm{d}W_t$.

Proof. F martingale $\Rightarrow \mathbb{E}_t \int_t^T \dot{Q}_s F_s\,\mathrm{d}s = S_t F_t$. In the differential, $\dot{Q}_t F_t\,\mathrm{d}t$ cancels with $\dot{S}_t F_t\,\mathrm{d}t$.

- $\Phi_t = \psi_t - S_t F_t$: a martingale (the **field observable**) minus the drift still expected on the unresolved scales.
- Noise of ψ: fresh shell fluctuation + **revision of the forecast**. The forward–backward coupling moves from the drift to the volatility.

ψ is the **interaction picture** w.r.t. the classical dynamics.

The FBSDE in forecast coordinates

Invert $x \mapsto \psi = x + S_t F_t(x)$: $x = \psi + S_t F_t(\psi)$, $F_t(\psi) := -F_t(x)$. Response and regulator:

$$H_t(\psi) = (S_t^{-1} + DF_t(\psi))^{-1}, \qquad R_t = S_t^{-1} - Q_T^{-1}, \qquad \dot{R}_t = S_t^{-1}\dot{Q}_t S_t^{-1} \geq 0.$$

Theorem. (formal)

$$\mathrm{d}\psi_t = H_t(\psi_t)\; S_t^{-1}\dot{Q}_t^{1/2}\,\mathrm{d}W_t, \qquad F_t(\psi_t) = \mathbb{E}_t\left[D V_T(\psi_T)\right],$$

$\psi_T = \Phi_T$. ***No drift***; *covariance* $\mathscr{C}_t = H_t \dot{R}_t H_t^*$. *The decoupling field solves the* ***pure diffusion*** *equation*

$$\partial_t F_t + \tfrac{1}{2}\mathsf{Tr}(\mathscr{C}_t\, D^2 F_t) = 0, \qquad F_T = D V_T.$$

Proof. $1 + S_t DF_t = H_t S_t^{-1}$; $F_t(\psi_t)$ is a martingale and ψ has no drift: Itô.

What the change of coordinates did

Compare with Burgers:

$$\underbrace{\partial_t F = -DF[\dot{Q}F] - \tfrac{1}{2}\mathrm{Tr}(\dot{Q}D^2F)}_{\text{field coordinate}} \quad \longrightarrow \quad \underbrace{\partial_t F = -\tfrac{1}{2}\mathrm{Tr}(\mathscr{C}\ D^2F)}_{\text{forecast coordinate}}$$

- The transport term is **absorbed exactly**: dropping the Itô term, $F_t = DV_T$ does not move. All **trees** live in the coordinate change $x = \psi + S_t F_t(\psi)$.
- **Loops** come only from the quadratic variation $\mathscr{C}_t$ of the forecast; the nonlinearity has moved into its field dependence.
- No gradient structure, no Gaussian convolution used: valid for any FBSDE with a Markov martingale force, whenever the coordinate change is invertible.

Conditional moments

Proposition. (formal) *If* $F_t = -D\,V_t$, *then*

$$H_t(\psi_t) = \mathrm{Cov}_t := \mathrm{Cov}^{\mathbb{P}}(\Phi_T \mid \mathscr{F}_t) = S_t - S_t D^2 V_t(\Phi_t)\, S_t.$$

So $H_t > 0$, $1 + S_t D F_t$ *is always invertible, and* $H_t \le S_t$ *for convex* V_T *(Brascamp–Lieb).*

The forecast coordinates are the first two conditional moments of the stochastic localisation: $(\psi_t, \mathrm{Cov}_t) = (\mathbb{E}_t \Phi_T, \mathrm{Cov}_t \Phi_T)$. Since $\mathbb{E}_t[\Phi_T \otimes \Phi_T] = \psi_t \otimes \psi_t + \mathrm{Cov}_t$ is a martingale,

$$\mathrm{d}\langle \psi, \psi \rangle_t = \mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t \,\mathrm{d}t, \qquad \mathrm{dCov}_t = -\mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t \,\mathrm{d}t + \mathrm{d}(\mathrm{mart.}).$$

Gaussian reference: $S_t \dot{R}_t S_t = \dot{Q}_t$.

Brascamp–Lieb bounds on the conditional covariance

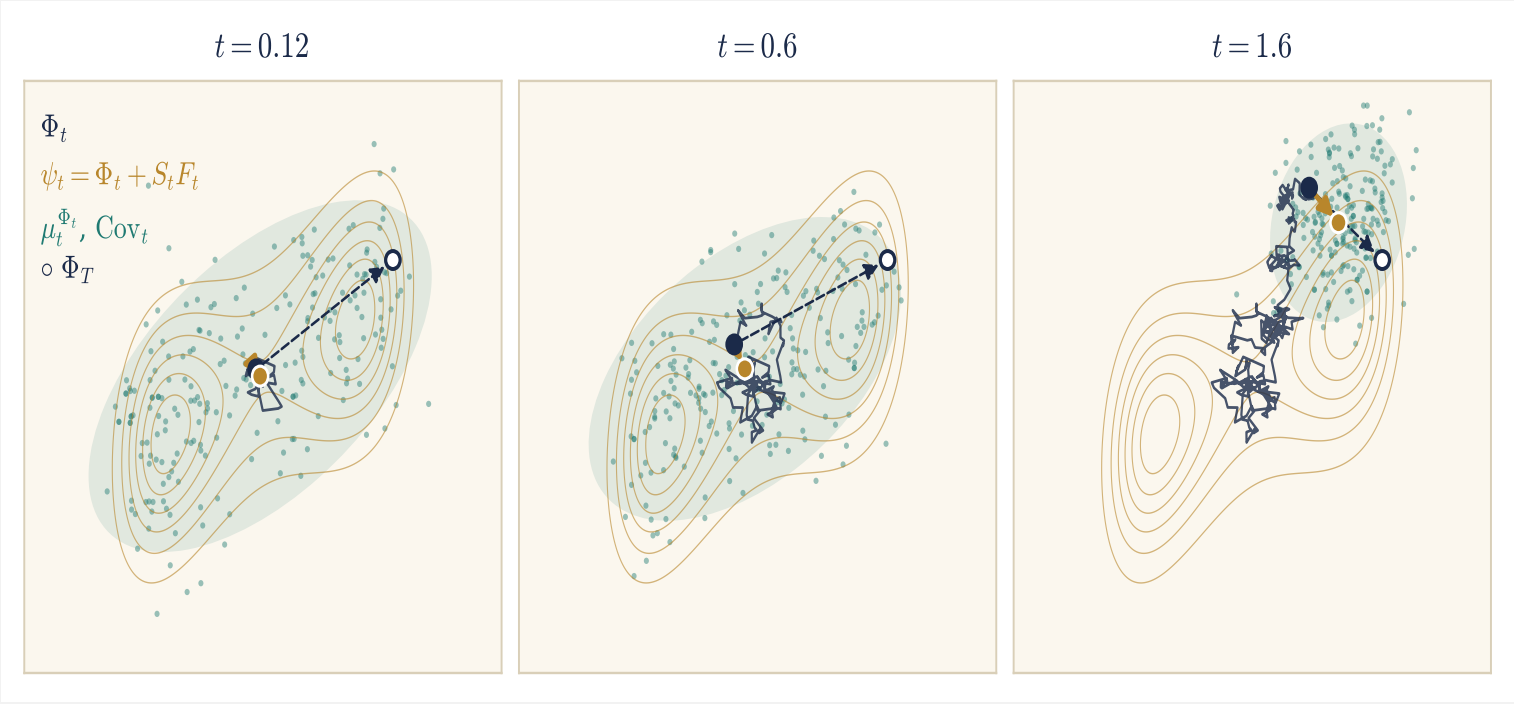
Conditional law $\mu_t^\phi \propto \mathrm{e}^{-U}$, $D^2 U = S_t^{-1} + D^2 V_T$. Then

$$(\mathbb{E}_t[S_t^{-1} + D^2 V_T])^{-1} \le \mathrm{Cov}_t \le \mathbb{E}_t[(S_t^{-1} + D^2 V_T)^{-1}].$$

- **Lower:** Cramér–Rao, no convexity needed: $\mathbb{E}[(y-m) \otimes D U] = 1$ and Cauchy–Schwarz.
- **Upper:** Brascamp–Lieb (1976), for $D^2 U > 0$: $\mathrm{Var}(f) \le \mathbb{E}\langle Df, (D^2 U)^{-1} Df\rangle$.
- Equal for quadratic V_T. Via $\mathrm{Cov}_t = S_t - S_t D^2 V_t S_t$: two-sided bounds on $D^2 V_t$, i.e. on the Wetterich propagator H_t.
- V_T convex $\Rightarrow \mathrm{Cov}_t \le S_t$, $D^2 V_t \ge 0$: convexity propagates along Polchinski. $D^2 V_T \ge \kappa \Rightarrow \mathrm{Cov}_t \le (S_t^{-1} + \kappa)^{-1}$.

Φ^4_3: Wick ordering gives V_T the curvature $-12\lambda c_T \to -\infty$ at $\varphi = 0$ (already for Φ^4_2). The upper bound is lost, the conditional law may spread over several wells: the **large-field problem** (clipping and global coercivity, Lecture 5).

Stochastic localisation: a picture



Two-component field, V_T a tilted double well (contours: ν_T). Path Φ_t, forecast $\psi_t = \Phi_t + S_t F_t$, conditional law $\mu_t^{\Phi_t}$ and 2σ ellipse of Cov_t; open circle: Φ_T. The conditional law localises on one well.

Phase space and Lagrangian graphs

Phase space of pairs (field, force) $(x,p) \in \mathbb{R}^n \times \mathbb{R}^n$, symplectic form $\omega = \sum_i \mathrm{d}x_i \wedge \mathrm{d}p_i$.

- L (dimension n) is **Lagrangian** if $\omega|_L = 0$.
- A graph $p = F(x)$ is Lagrangian $\Leftrightarrow$ DF symmetric $\Leftrightarrow$ F is a gradient. **Gradient sector = Lagrangian graphs.**
- Symplectic maps send Lagrangian submanifolds to Lagrangian ones. The shear $\Sigma_t(x,p) = (x + S_t p, p)$ is symplectic since S_t is symmetric: $\Sigma_t^* \omega = \omega + \sum S_{ij} \mathrm{d}p_j \wedge \mathrm{d}p_i = \omega$.

The image of the graph of $-DV_t$ is Lagrangian; as a graph over the forecast ψ it is the graph of $-D\Gamma_t$, with the Moreau transform as generating function.

The gradient sector

$F_t = -D\,V_t$: the graph $\{(x, -D\,V_t(x))\}$ is a **Lagrangian submanifold** of (field, force) space. The shear Σ_t is symplectic (S_t symmetric), and the image is a graph over ψ ($x \mapsto x + S_t F_t(x)$ is $S_t\,D$ of a strictly convex function). Hence

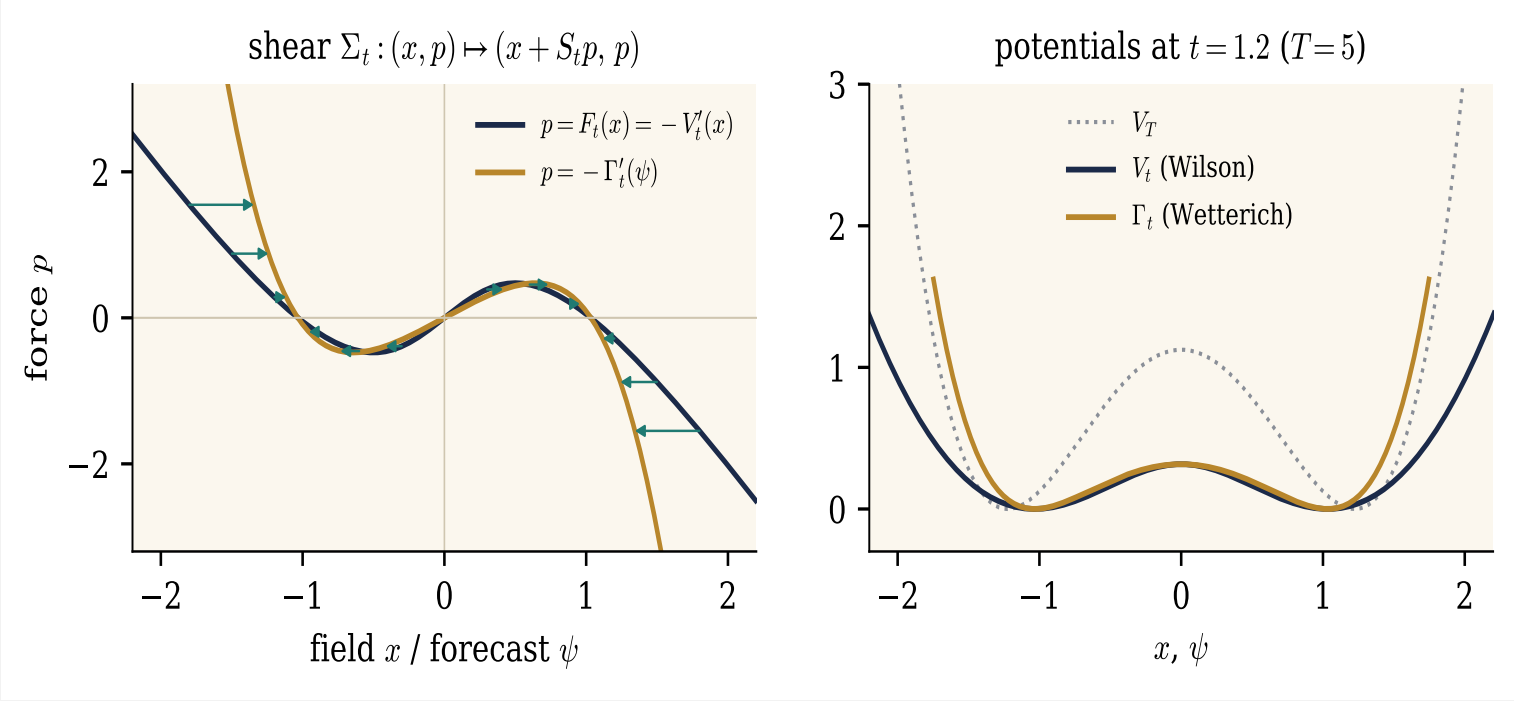
$$F_t = D\,\Gamma_t, \qquad H_t = (S_t^{-1} + D^2\Gamma_t)^{-1}, \qquad \Gamma_T = V_T.$$

Generating function: the **Moreau transform**

$$V_t(x) = \inf_{\psi} \left\{ \Gamma_t(\psi) + \tfrac{1}{2}\langle x - \psi, S_t^{-1}\,(x - \psi)\rangle \right\},$$

optimality: $x = \psi + S_t\,D\,\Gamma_t(\psi)$, $D\,V_t(x) = D\,\Gamma_t(\psi)$. Well defined: $\frac{1}{2}\,\langle x, S_t^{-1}\,x\rangle - V_t(x)$ has Hessian $S_t^{-1}\mathrm{Cov}_t S_t^{-1} > 0$.

The shear in zero dimensions



Left: Σ_t maps the graph of $F_t = -V_t'$ to the graph of $-\Gamma_t'$ over the forecast.
Right: V_T, V_t and Γ_t ($t = 1.2$, $T = 5$).

The Wetterich equation

Theorem. (formal)

$$\partial_t \Gamma_t = \tfrac{1}{2}\mathrm{Tr}\big[\dot{R}_t\,((S_t^{-1} + D^2\Gamma_t)^{-1} - S_t)\big], \qquad \Gamma_T = V_T.$$

Equivalently, for the ***effective average action*** $\Gamma_t^{\mathrm{EAA}} = \Gamma_t + \frac{1}{2}\langle \psi, Q_T^{-1}\psi\rangle$,

$$\partial_t \Gamma_t^{\mathrm{EAA}} = \tfrac{1}{2}\mathrm{Tr}\big[\dot{R}_t\,(D^2\Gamma_t^{\mathrm{EAA}} + R_t)^{-1}\big] + \textit{const},$$

from the bare action ($t = T$, $R_T = \infty$) to the 1PI effective action of ν_T ($t = 0$, $R_0 = 0$).

Proof. $D_v \mathrm{Tr}(\dot{R}H) = -\mathrm{Tr}(\mathscr{C}\ D^2 F)[v]$ by symmetry of $D^3\Gamma$, so the force equation reads $D\left(\partial_t\Gamma - \frac{1}{2}\mathrm{Tr}(\dot{R}H)\right) = 0$. The constant: envelope theorem + Polchinski.

Wetterich = the backward half of the FBSDE

Every object in the Wetterich equation is an observable of the Polchinski process.

Wetterich	FBSDE
field argument ψ	forecast $\psi_t = \mathbb{E}_t[\Phi_T]$
$D\Gamma_t(\psi_t)$	minus the costate, $\mathbb{E}_t[D V_T(\Phi_T)]$
propagator $(D^2\Gamma_t^{\mathsf{EAA}} + R_t)^{-1}$	conditional covariance Cov_t
regulator $R_t = S_t^{-1} - Q_T^{-1}$	precision of the unresolved shell minus the prior's
$\partial_t \Gamma_t$	$\frac{1}{2}\mathrm{Tr}[\dot{R}_t\,(\mathsf{Cov}_t - S_t)]$

Driven by **interacting minus free** conditional covariance; **one-loop exact**, with the full propagator; in the Gaussian case only the constant flows. Wetterich 1993; Morris 1994; Berges–Tetradis–Wetterich 2002

The regulator of the heat profile

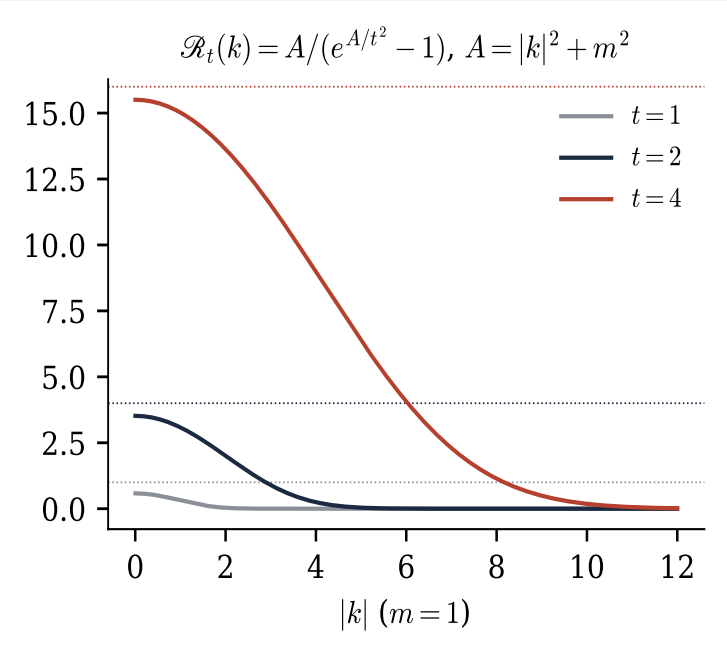
For $T=\infty$, $S_t = C\,(1-\mathrm{e}^{-A/t^2})$ and

$$R_t = \frac{A}{\mathrm{e}^{A/t^2}-1}.$$

- $\approx t^2$ (a mass term) on the modes $|k| \ll t$;
- exponentially small on $|k| \gg t$.

The **exponential regulator** of the functional RG.

Other profiles give other regulators, e.g. the mass regulator $R_t = t^2$.



Variational meaning and the tree level

Constrained free energy. Gibbs variational principle for the conditional law:

$$\Gamma_t(\psi) = \inf\left\{\int V_T \,\mathrm{d}\nu + \mathsf{Ent}(\nu \,|\, \mu^{S_t}(\cdot - \psi)) \colon \int \varphi \,\mathrm{d}\nu = \psi\right\}\colon$$

the free energy of the unresolved shell, **constrained to the forecast**. V_t is the value function; Γ_t is its Moreau transform.

Tree level. Drop the Itô term: $\Gamma_t = V_T$, and the Moreau transform becomes the **Hopf–Lax formula**

$$V_t^{\mathsf{tree}}(x) = \inf_y \left\{ V_T(y) + \tfrac{1}{2}\langle y - x, S_t^{-1}(y-x)\rangle \right\},$$

which solves $\partial_t V = \frac{1}{2}\langle DV, \dot{Q} DV\rangle$: the saddle point of the Gaussian convolution. The minimiser solves the classical field equation $S_t^{-1}(y-x) + DV_T(y) = 0$ in the shell $[t,T]$; expanding in V_T gives exactly the tree diagrams.

Entropy along the localisation

$\rho \geq 0$, $\mathbb{E}_{\nu_T} \rho = 1$, $M_t = \mathbb{E}_t \rho(\Phi_T) = G_t(\Phi_t)$, $G_t(\phi) = \int \rho \, \mathrm{d}\mu_t^\phi$.

- $M_0 = 1$, $M_T = \rho(\Phi_T)$, and $\mathrm{d}M_t = \langle D G_t(\Phi_t), \dot{Q}_t^{1/2} \mathrm{d}W_t \rangle$.
- Itô for $M \log M$:

$$\mathrm{Ent}_{\nu_T}(\rho) = \frac{1}{2} \mathbb{E} \int_0^T \frac{\mathrm{d}\langle M \rangle_t}{M_t} = 2 \, \mathbb{E} \int_0^T \left\| \dot{Q}_t^{1/2} D \sqrt{G_t}(\Phi_t) \right\|^2 \mathrm{d}t.$$

- Stopping at t: **revealed + hidden** entropy,

$$\mathrm{Ent}_{\nu_T}(\rho) = \mathrm{Ent}(M_t) + \mathbb{E} \, \mathrm{Ent}_{\mu_t^{\Phi_t}}(\rho).$$

- The increments see the conditional law: $D G_t(\phi) = S_t^{-1} \mathrm{Cov}_{\mu_t^\phi}(\rho, y)$.

Bauerschmidt–Bodineau–Dagallier, Probab. Surveys 2024: their fluctuation measure is our stochastic localisation (scale reversed).

Multiscale Bakry–Émery

Theorem (Bauerschmidt–Bodineau). If for all t, ϕ, with $\kappa_t \in \mathbb{R}$ (possibly negative),

$$\dot{Q}_t D^2 V_t(\phi)\, \dot{Q}_t + \frac{1}{2}\ddot{Q}_t \geq \kappa_t \dot{Q}_t,$$

then $\mathrm{Ent}_{\nu_T}(\rho) \leq \frac{2}{\gamma}\mathbb{E}\|\dot{Q}_T^{1/2} D\sqrt{\rho}\|^2$, $\frac{1}{\gamma} = \int_0^T \mathrm{e}^{-2\int_t^T \kappa}\,\mathrm{d}t$.

- Gradient bound at each scale + entropy formula. Gaussian check: $V = 0$, $Q_t = t\,Q_T/T$, $\kappa = 0 \Rightarrow$ Gaussian LSI.
- Convexity of the **effective** potentials, not of V_T.
- By $D^2 V_t = S_t^{-1} - S_t^{-1}\mathrm{Cov}_t\, S_t^{-1}$ and $\dot{Q} = S\dot{R}S$, the hypothesis is the bound, uniform in ϕ,

$$\mathrm{Cov}_t \leq S_t - \kappa_t \dot{R}_t^{-1} + \frac{1}{2}\dot{R}_t^{-1} S_t^{-1} \ddot{Q}_t S_t^{-1} \dot{R}_t^{-1}$$

  ($= S_t$ for the Gaussian): $\kappa_t > 0$ is more concentrated than the unresolved Gaussian.
- **Entropic stability** (Chen–Eldan): $\dot{R}_r\,\mathrm{Cov}_{\mu_r^\phi}\dot{R}_r \leq \alpha_r \dot{R}_r$ for all ϕ, $r \leq t \Rightarrow \mathrm{Ent}(\rho) \leq \mathrm{e}^{\int_0^t \alpha}\,\mathbb{E}\,\mathrm{Ent}_{\mu_t}(\rho)$: the low resolutions must not hide too much.

Variational structure

Lecture 4

The Boué–Dupuis formula

$\mathbb{H}_a$: adapted drifts with $\mathbb{E}\int_0^T \|u_s\|^2 < \infty$; $X^u = X + \int_0^\cdot \dot{Q}_s^{1/2} u_s \, \mathrm{d}s$.

Theorem. (Boué–Dupuis 1998; Polchinski = HJB) (formal)

$$-\log \mathbb{E}_0[\mathrm{e}^{-V_T(X_T)}] = V_0(0) = \inf_{u \in \mathbb{H}_a} \mathbb{E}_0\left[V_T(X_T^u) + \frac{1}{2}\int_0^T \|u_s\|^2 \mathrm{d}s \right].$$

Optimal feedback $u_s^* = -\dot{Q}_s^{1/2} D\, V_s(X_s^{u^*})$, $\mathrm{Law}(X_T^{u^*}) = \nu_T$. *$V_t$ is the* ***value function****; Polchinski is the* ***HJB equation****.*

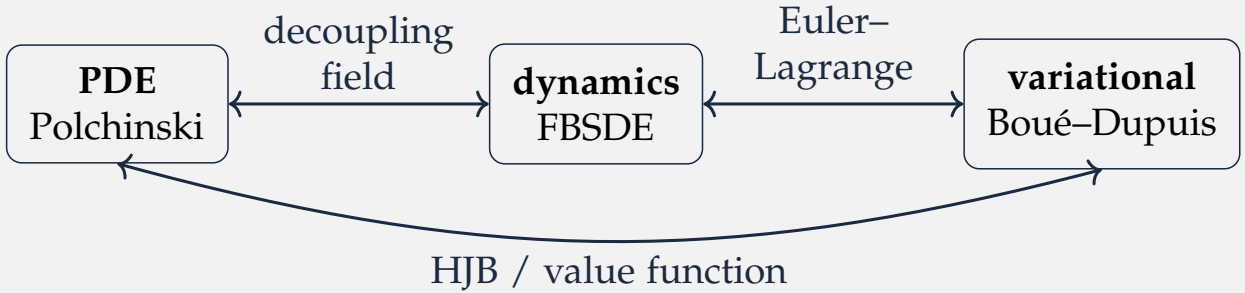
Proof (verification). Itô + Polchinski:

$$\mathbb{E}_0\left[V_T(X_T^u) + \tfrac{1}{2}\int_0^T \|u_t\|^2 \right] = V_0(0) + \tfrac{1}{2}\mathbb{E}_0 \int_0^T \|u_t + \dot{Q}_t^{1/2} D\, V_t(X_t^u)\|^2 \mathrm{d}t.$$

Three faces of one object

Euler–Lagrange. Perturb $u \to u + \varepsilon v$:

$$u_s = -\dot{Q}_s^{1/2}\, \mathbb{E}_s\left[D V_T(X_T^u)\right] = \dot{Q}_s^{1/2} F_s: \quad \text{the FBSDE.}$$



Entropy. $\frac{1}{2}\mathbb{E}_0 \int \|u\|^2 \geq \mathsf{Ent}(\mathsf{Law}(X^u)|\mathsf{Law}(X))$: a pathwise lift of the Gibbs principle $-\log Z_T = \inf_\nu \{\int V_T \, d\nu + \mathsf{Ent}(\nu|\mu^{Q_T})\}$. Useful because adapted drifts can be **built scale by scale**, as functionals of X.

Renormalisation in the variational picture

Generating functional:

$$\mathscr{W}_T(f) = -\log \int \mathrm{e}^{-f} \mathrm{d}\nu_T = \inf_u \mathbb{E}_0 \Big[(f + V_T)(X_T^u) + \frac{1}{2} \int_0^T \|u_s\|^2 \Big] - V_0(0).$$

Bounds uniform in $T \Rightarrow$ tightness, moments, identification of limits.

Φ_3^4. The leading drift $u_s \approx -4\lambda \dot{Q}_s^{1/2} {:}X_s^3{:}$ has **divergent cost**,

$$\tfrac{1}{2}\mathbb{E} \int_0^T \| 4\lambda \dot{Q}_s^{1/2} {:}X_s^3{:} \|^2 \mathrm{d}s = 12\lambda^2 L^3 \int Q_T(z)^4 \, \mathrm{d}z \sim T.$$

Change variables $u = -4\lambda \dot{Q}^{1/2} {:}X^3{:} + r$, then paracontrolled changes of variables: the singular terms become explicit polynomials in Wick powers of X, **cancelled by the counterterms**, and the remainder is coercive. Uniform control in Besov/paracontrolled norms = construction of Φ_3^4. Barashkov–Gubinelli, Duke 2020; EJP 2021

Infinite volume

On $\mathbb{T}_L^d$, $V_0(0)$ and V_t are **extensive** ($\sim L^d$). Three **intensive** objects survive:

i. **the force.** For $V_T = \int v_T(\varphi)$, on $\mathbb{R}^d$:

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t(x) = -\mathbb{E}_t\big[v_T'(\Phi_T(x))\big].$$

Everything is local; infinite-volume limits should solve this. Uniqueness of solutions $\leftrightarrow$ uniqueness of the theory; non-uniqueness expected at phase transitions (Φ_2^4, deep double well).

ii. **the Burgers flow** of the force;

iii. **differences of free energies** $\mathscr{W}_T^{(L)}(f)$, f local: bounded as $L \to \infty$; basis of the infinite-volume analysis of Φ_2^4 and $\exp(\Phi)_2$.

Estimates must be **local** (weighted norms, volume-independent constants): the analogue of "coming down from infinity". Barashkov–Gubinelli, PMP 2023; Glimm–Jaffe–Spencer 1975

Differences of free energies

Perturb the optimal drift: $u = u^* + w$, $X^u = \Phi + I(w)$, $I_t(w) = \int_0^t \dot{Q}^{1/2} w$. Euler–Lagrange gives the cross term

$$\mathbb{E} \int \langle u^*, w \rangle = -\mathbb{E} \langle D V_T(\Phi_T), I_T(w) \rangle,$$

and the two extensive infima cancel exactly:

$$-\log \int \mathrm{e}^{-f} \, \mathrm{d}\nu_T = \inf_w \mathbb{E} \left[f(\Phi_T + I_T(w)) + B_{V_T}(\Phi_T, I_T(w)) + \frac{1}{2} \int_0^T \|w\|^2 \right],$$

$B_V(\phi, h) = V(\phi + h) - V(\phi) - \langle D V(\phi), h \rangle$: the **first-order Taylor remainder**.

- Minimiser: $\Phi^f = \Phi + I(w)$ solves the FBSDE with terminal potential $V_T + f$, a perturbation of Φ: $F^f - F = -\mathbb{E}_s[Df + D V_T(\Phi_T^f) - D V_T(\Phi_T)]$.

Infinite-volume measures

Local interaction $V_T = \int v_T(\varphi(x))\,\mathrm{d}x$:

$$B_{V_T}(\phi, h) = \int b_T(\phi(x), h(x))\,\mathrm{d}x, \qquad b_T(a,h) = v_T(a+h) - v_T(a) - v_T'(a)\,h.$$

- $b_T(a,0) = 0$: only the region where the perturbation acts counts. Every term is **intensive**: no free energy, only Φ and its perturbation.
- f local $\Rightarrow$ the optimal w should decay away from its support, at the rate of decay of correlations.
- Several solutions of the FBSDE (phase transitions) $\Rightarrow$ several measures, each with its own principle.
- Φ^4_3: the remainder term carries the counterterms; the change of variables of Barashkov–Gubinelli is still needed.

Formal statement. An infinite-volume measure is the law of Φ_T for a solution of the infinite-volume FBSDE satisfying the relative variational principle for all local f.

Additional material

Lecture 4

A semigroup of renormalisation maps

$\mathcal{R}_S(V) = -\log\,(\mu^S * \mathrm{e}^{-V})$. Since $\mu^S * \mu^{S'} = \mu^{S+S'}$,

$$\mathcal{R}_S \circ \mathcal{R}_{S'} = \mathcal{R}_{S+S'} = \mathcal{R}_{S'} \circ \mathcal{R}_S, \qquad V_t = \mathcal{R}_{Q_{T,t}}(V_T):$$

an orbit of an **abelian** semigroup indexed by the cone of positive covariances, nonlinear on V, linear on e^{-V}. Generators:

$$\mathcal{P}_B(V) = \tfrac{1}{2}\mathrm{Tr}(B\,D^2 V) - \tfrac{1}{2}\langle D\,V, B\,D\,V\rangle, \qquad \partial_t V_t = -\mathcal{P}_{\dot{Q}_t}(V_t).$$

Proposition. (Flatness) (formal) $[\mathcal{P}_B, \mathcal{P}_{B'}] = 0$. *$V_t$ depends on the profile only through $Q_{T,t}$; $V_0(0)$ and ν_T only through Q_T.*

Different profiles with the same endpoints are different **coordinate systems** on the same RG trajectory: they change Φ and intermediate observables, not the terminal theory. **No holonomy.**

Perturbation theory and counterterms

Gaussian convolution **lowers the Wick ordering**: $V_t = \int [\lambda :\phi^4:_t + \gamma_T :\phi^2:_t] + O(\lambda^2)$, the dual of $\int H_4(X_t; c_t)$ being an observable. At second order, $S = Q_{T,t}$:

$$V_t = \int [\lambda :\phi^4:_t + \gamma_T :\phi^2:_t] - \frac{\lambda^2}{2} \sum_{k=1}^{4} \binom{4}{k}^2 k! \iint S(x,y)^k :\phi^{4-k}:_t (x) :\phi^{4-k}:_t (y) + O(\lambda^3).$$

$T \to \infty$ at fixed t, $d = 3$, $G(z) \sim (4\pi|z|)^{-1}$:

- $k = 1, 2$: finite, nonlocal.
- $k = 3$ (**sunset**): $S^3 \sim |z|^{-3}$; local divergence $-48\lambda^2 (\int S^3) \int :\phi^2:_t$ (+ const), cancelled by $\gamma_T = 48\lambda^2 \int Q_T^3 = \frac{3\lambda^2}{\pi^2} \log T + O(1)$.
- $k = 4$: vacuum energy $\sim T$; cancels in ν_T.

Renormalisation = choosing V_T so that V_t at **fixed** resolution converges. $\nu_t \ll \mu^{Q_t}$ expected for $t < \infty$; the terminal Φ^4_3 is singular w.r.t. the GFF.

Dyson–Schwinger equations at every scale

Operators functions of A; $K = Q_T^{-1}$, $J_t = K Q_t$ (heat profile: a heat semigroup, $\to e^{-A/t^2}$).

Theorem. (formal) *The **noise observable** $\eta_t = K\Phi_t - J_t F_t$ is a centred martingale,* $\mathrm{d}\eta_t = K\dot{Q}_t^{1/2}\mathrm{d}W_t^{\mathbb{P}} - J_t\,\mathrm{d}F_t$, *and*

$$\mathbb{E}^{\mathbb{P}}\left[D_{J_t h} G(\Phi_t)\right] = \mathbb{E}^{\mathbb{P}}\left[G(\Phi_t)\,\langle h, K\Phi_t - J_t F_t\rangle\right].$$

At $t = T$: the DS equation $\mathbb{E}^{\nu_T}[D_h G] = \mathbb{E}^{\nu_T}[G\langle h, K\varphi + D V_T\rangle]$.

Proof. $\dot{J}_t = K\dot{Q}_t$: drifts cancel. IBP for μ^{Q_t} in direction $J_t h$, using $\Phi_t \sim \nu_t$.

The field equation at resolution t

As $T \to \infty$, $K \to A$, and $\eta_t = K\Phi_t - J_t F_t$ becomes

$$A\Phi_t = AQ_t F_t + \eta_t \qquad (AQ_t = \mathrm{e}^{-A/t^2} \text{ for the heat profile}):$$

the Euclidean field equation $A\varphi = -V'(\varphi) + \text{noise}$, **seen at resolution t**. No products of distributions: the nonlinearity enters only through the force **observable**, smeared at scale $1/t$. For Φ^4_3, with F_t in the role of the renormalised cube, $t \to \infty$ corresponds to the renormalised DS equation (Gubinelli–Hofmanová 2021).

Compare: stochastic quantisation (Lecture 3)

There the field equation is imposed in an **auxiliary time**; here it holds at **each resolution**, and the auxiliary parameter is the resolution itself.

Conditioning an SPDE solution on a scale filtration gives identities of the same shape but with a different noise: the martingale $J_t\,\mathrm{d}F_t$ in $\mathrm{d}\eta_t$ is essential.

Summary

- Interactions: effective potentials V_t solve **Polchinski**; the interacting field solves an **FBSDE** whose force is an observable.
- In the forecast coordinate $\psi_t = \mathbb{E}_t \Phi_T$ the FBSDE has no drift, trees are integrated exactly, and the force flow is pure diffusion; in the gradient case it is **Wetterich's** equation.
- PDE (Polchinski) $\leftrightarrow$ dynamics (FBSDE) $\leftrightarrow$ variational principle (Boué–Dupuis).
- Counterterms and DS equations all live at **finite resolution**.

Next

Lectures 5–7: a construction of Φ^4_3 on the torus via the FBSDE.

Exercises and reading

Exercises (numbering of the notes)

- 7.1 Quadratic interactions; 7.2 zero dimensions.
- 8.1 Stochastic maximum principle; 8.2 characteristics.
- 8.5 The constant; 8.7 1PI effective action.
- 8.8 Trees versus loops; 8.9 regulators.
- 9.2 Cost of the leading drift.
- 10.2 The sunset; 10.3 power counting.
- 11.1 DS from the dynamics.

Reading

- Bauerschmidt–Bodineau–Dagallier, *Stochastic dynamics and the Polchinski equation*, Probab. Surveys 2024.
- Barashkov–Gubinelli, *A variational method for* Φ^4_3, Duke 2020.
- Berges–Tetradis–Wetterich, Phys. Rep. 2002.
- Ma–Yong, *FBSDEs and their applications*, LNM 1702.